15. Give the uses of argon.
16. Write the valence shell electronic configuration of group-15 elements.
17. Give two equations to illustrate the chemical behaviour of phosphine.
18. Give a reaction between nitric acid and a basic oxide.
19. What happens when PCl_5 is heated?
20. Suggest a reason why HF is a weak acid, whereas binary acids of the all other halogens are strong acids.
21. Deduce the oxidation number of oxygen in hypofluorous acid – HOF.
22. What type of hybridisation occur in

    a) BrF_5 b) BrF_3

23. Complete the following reactions.

    1. $NaCl + MnO_2 + H_2SO_4 \longrightarrow$
    2. $NaNO_2 + HCl \longrightarrow$
    3. $P_4 + NaOH + H_2O \longrightarrow$
    4. $AgNO_3 + PH_3 \longrightarrow$
    5. $Mg + HNO_3 \longrightarrow$
    6. $KClO_3 \xrightarrow{\Delta}$
    7. $Cu + \overset{\text{Hot conc}}{H_2SO_4} \longrightarrow$
    8. $Sb + Cl_2 \longrightarrow$
    9. $HBr + H_2SO_4 \longrightarrow$
    10. $XeF_6 + H_2O \longrightarrow$
    11. $XeO_6^{4-} + Mn^{2+} + H^+ \longrightarrow$
    12. $XeOF_4 + SiO_2 \longrightarrow$
    13. $Xe + F_2 \xrightarrow[400^0C]{\text{Ni / 200 atm}}$

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